



Distributed Systems and Internet technologies

PERSISTENCE

Juan Luis Gorricho

PERSISTENCE

- There are three persistence mechanisms:
  - Preferences
  - Files:
    - Internal files (phone memory)
    - External files (external memory)
  - Data bases
- The information is not shared among different apps, except for external files.

PREFERENCES

- Tool to save pairs of key-value using native Java types for the values.
- Preferences with Application scope: **getSharedPreferences**(fileName,mode); saved at: /data/data/package_name/shared_prefs
- Preferences with Activity scope: **getPreferences**(mode); it takes the name of the activity for the saved file.

PREFERENCES

- To write or modify a pair key-value you must first retrieve an **editor**: SharedPreferences.Editor
- To read a preference use the get() method.

INTERNAL FILES

- Two operational directories are defined:
  - Internal directory: context.getFilesDir();
  - Cache directory: context.getCacheDir();
- To create a file:
  - File file = new File(context.getFilesDir(), filename);
  - File file = File.createTempFile(prefix, sufix, context.getCacheDir());
  - Alternatively, for the internal directory:
    - FileInputStream fis = context.openFileInput(filename);
    - FileOutputStream fos = context.openFileOutput(filename);

INTERNAL FILES

- Alternative to delete a file from the internal directory:
  - context.deleteFile(fileName);
- The internal files are stored in the internal memory of the cell-phone at:
  - /data/data/package_name/files
  - /data/data/package_name/cache

EXTERNAL FILES

- To check if the external memory is available:

```
String state = Environment.getExternalStorageState();
if (Environment.MEDIA_MOUNTED.equals(state)) { Okay! }
```

- Two categories of files (but all are public) are defined:
  - Public files, are not deleted when uninstalling the app.
  - Private files, are deleted when the app is uninstalled.

EXTERNAL FILES

- To work with the two types of files make use of the directories:

  Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_PICTURES);

  context.getExternalFilesDir(null);

- The file access is conventional Java coding:

  File file = new File(dir, filename);

- The private files are saved at:

  /mnt/sdcard/Android/data/package_name/files

EXTERNAL FILES

- It is recommended to use the subdirectories predefined by Environment.
- To check the available space at the external memory: dir.getFreeSpace() and the total space: dir.getTotalSpace()

PERSISTENCE

- Read only files from "resources":
  - InputStream is = resources.openRawResource(R.raw.mydatafile); located at res/raw/mydatafile
  - InputStream is = context.getAssets().open(filename); located at Assets
- Data bases:
  - Android supports SQLite
  - The **SQLiteDatabase** class implements the ddbb access logic.

EXERCISE

- Review the programming of the methods contained in the GlobalState java class using the "internal files" option to implement the persistence.
- Modify the **MessagesActivity** to locally store the downloaded messages; and to only request for new messages from the remote server.

EXERCISE

- Modify the WelcomeActivity to use the two buttons in a different way if a user identity is stored locally, so you will show this view instead:

